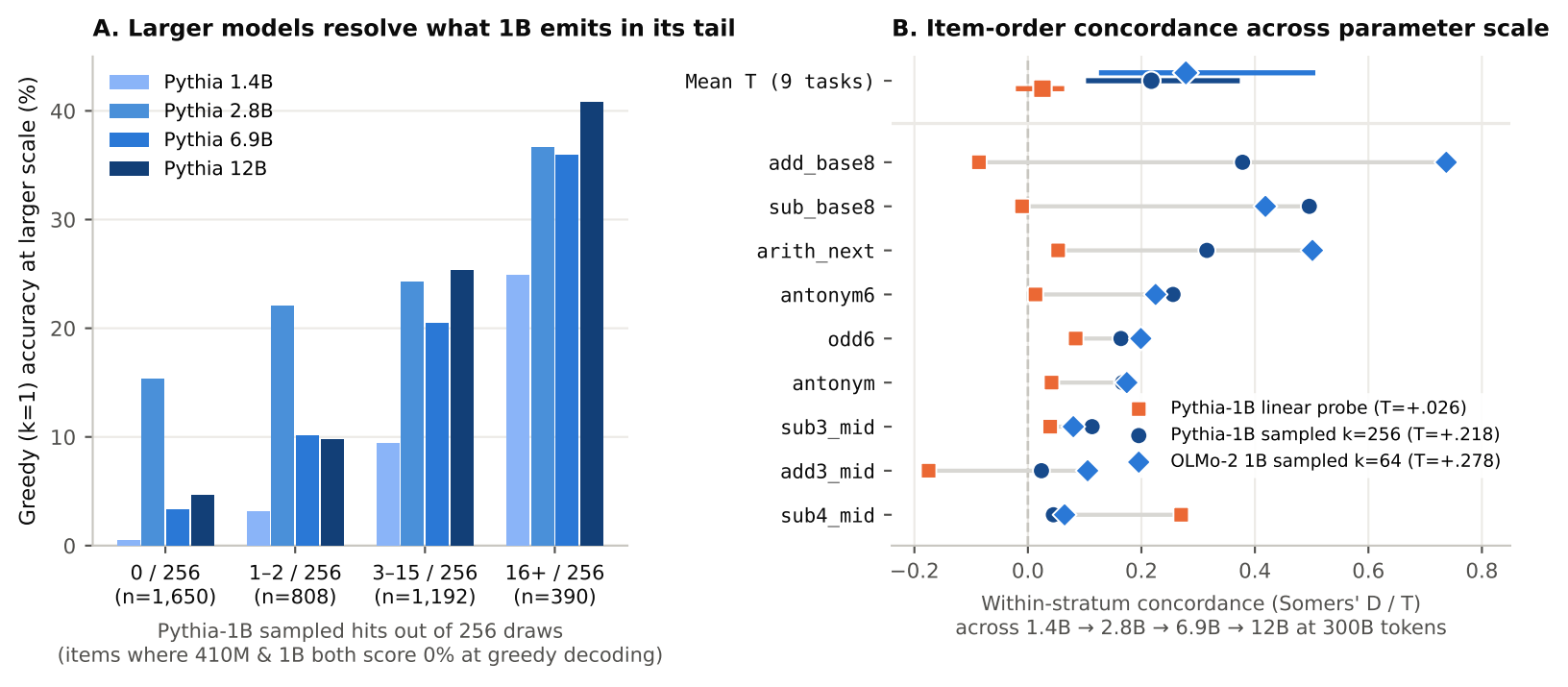
A. Larger models resolve what 1B emits in its tail
Pythia 1.4B
Pythia 2.8B
Pythia 6.9B
Pythia 12B
Greedy (k=1) accuracy at larger scale (%)
0
10
20
30
40
0 / 256
(n=1,650)
1–2 / 256
(n=808)
3–15 / 256
(n=1,192)
16+ / 256
(n=390)
Pythia-1B sampled hits out of 256 draws
(items where 410M & 1B both score 0% at greedy decoding)
B. Item-order concordance across parameter scale
Mean T (9 tasks)
add_base8
sub_base8
arith_next
antonym6
odd6
antonym
sub3_mid
add3_mid
sub4_mid
Pythia-1B linear probe (T=+.026)
Pythia-1B sampled k=256 (T=+.218)
OLMo-2 1B sampled k=64 (T=+.278)
−0.2
0.0
0.2
0.4
0.6
0.8
Within-stratum concordance (Somers' D / T)
across 1.4B → 2.8B → 6.9B → 12B at 300B tokens